



HELP Math 2.0 Scope

The purpose of this document is a follow-up to recent Istation and Boulder Learning meetings and conversations about how to fund HELP Math redevelopment/upgrade to V2.0. The HELP with English Language Proficiency Program (HELP Math or HELP 1.5) is a comprehensive, supplemental digital mathematics intervention designed specifically for use by English Learners and other struggling students, students with math learning disabilities and is also used as a comprehensive, supplemental remediation using the Response to Intervention Framework. Academic Language and vocabulary and the development of prerequisite knowledge and skills are fundamental cornerstones of the program. HELP Math is still today one of the only online, award winning instructional program in the U.S. to embed sheltered instruction and other research-based instructional strategies directly into a mathematics curriculum.

HELP Math helps students develop foundational and grade-level math knowledge and skills, and concurrently develop their academic vocabulary and overall English language abilities. The program has a flexible digital learning environment that applies Universal Design for Learning (UDL) principles ([CAST, 2011](#)). For example, HELP Math presents math concepts, procedures, and information in multiple ways, delivers content using audio and video, has a customizable learning platform enabling academic supports to be turned on or off according to each student's needs, explains academic and technical vocabulary and math symbols, models and provides guided practice, and is interactive with multiple and varied virtual tools for practice, expression and engagement. The key features that distinguish HELP Math from other supplemental programs used by ELLs is its evidence-based pedagogy that is embedded into the digital content; its flexible digital learning environment, and its use of adaptive technology and user path analysis to deliver engaging personalized instruction. Specifically, the intervention consists of the following components: (1) over 300 hours of interactive, standards-aligned math content; (2) a digital media assets library with virtual manipulatives (e.g., number lines, base 10 blocks, protractors, etc.); (3) diagnostic assessments, including pre-post tests used to place

students in the content and measure their achievement at the end of their curriculum, and formative assessments used to monitor students' progress toward reaching proficiency; (4) a customizable learning environment with interactive virtual support tools that can be accessed anytime and phased out when no longer needed (e.g., key terms dictionary, bilingual translations, digital writing journal, social network tools); and (5) a computational engine with an adaptive-prescriptive progress monitoring system that tracks students' interactions, dynamically sequences content to match student need, and provides continuous intelligent feedback. HELP Math content is highly interactive, colorful and musical, with interwoven language support. The program synchronizes audio, illustration, and text to create a visual connection between words and meaning (e.g., corresponding vocabulary, symbols or pictures flash in sync with audio). HELP Math engages students with the inherent need for cognitive stability in their development, and their need to feel supported as they move to their next level of competence (Vygotsky,1978).

Help Math (HM) has an outstanding interest, given years of sales in 28 states, an existing and historical customer base which can be revisited/expanded, and a proven digital pedagogy, to modernize and revise the current application as follows:

- Using HM V. 1.5 as “storyboards”- update and modernize the animation to state of the art with Istation animation team
- Technology conversion - Reprogram existing content/ Flash elements with HTML 5
- Grade Expansion- develop new K-2 Content through creation of additional lessons (approximately 27 K-2 lessons)
- Administration Enhancement - Student and Teacher usage console improvements
- Academic Vocabulary & Reading Practice Inclusion - Incorporate Boulder Learning speech recognition technology to enhance the program with reading practice through the curriculum technology and also spoken dialogs with a virtual tutor based upon decades of BLI research and millions in funding from NSF, IES and DE.
- Update - “Teacher Resources” alignment tables for TEKS, CCSS and specialized state standards for select target markets.
- Add additional language support

Basically, the scope of the project is as follows; to update HM v 1.5 animation to state of the art, update technology to HTML 5 and include grades K-2, reading practice, and virtual tutor technology, alignments and additional languages. As such, we estimate the following;

Estimated Needs

Full product enhancement review and design (to make sure all the Real World's and other content are fully current.

Project Management,

Animation Upgrade,

Flash to HTML5 conversion (approximately 300 hours of content);

Development of K-2 content (27 lessons @ \$5K/lesson)

Reading practice design development & Math Discourse R&D

Virtual Tutor Development, (BLI)

Q&A Testing,

Update admin and alignments

Additional language support,

Estimated timeframe

With funding in the range above, a six month product review and design phase followed by a 1 year development cycle 6 month Q&A testing is reasonable. In our view, it would be possible to continue selling HM 1.5 and “upsell” to V.2.0 once a full grade content is completed.

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